Task 4: Optimal Externalization Threshold

1. Validation PnL vs. Externalization Threshold (θ)

The externalization strategy forcibly closes an incoming trade with a third party if the model's predicted probability of adversity exceeds a threshold θ. Below is the validation PnL curve for $\tau = 30$, demonstrating the concave relationship between the chosen threshold and profitability.

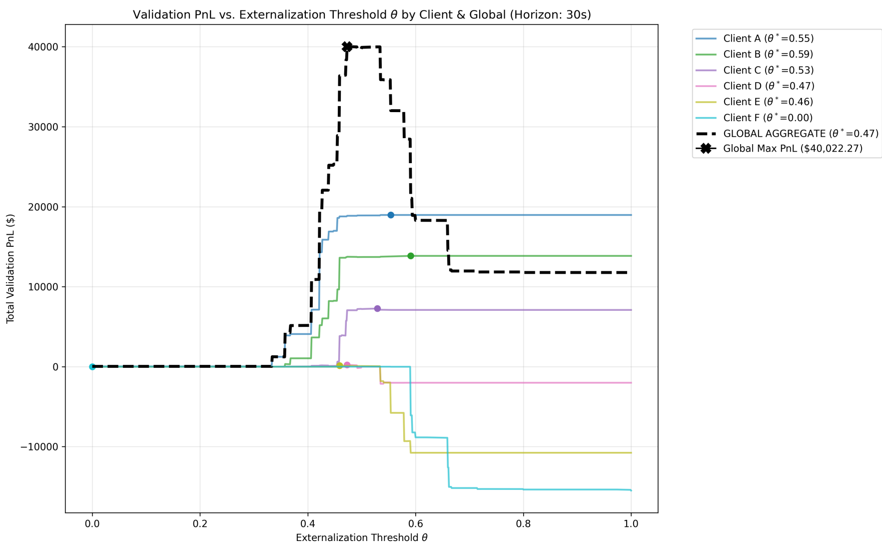

Figure 1: Validation PnL across $\theta \in [0, 1]$ for $\tau = 30$. The curve peaks at the optimal global threshold $\theta^* = 0.4730$.

2. The Trade-offs of Externalization

The parabolic shape of the validation curve perfectly illustrates the delicate balance of risk management in market making:

- **Over-Externalization (θ is too low):** When $\theta \to 0$, the Liquidity Provider (LP) becomes excessively risk-averse, externalizing almost all flow. Because we assume externalization nets exactly zero PnL, the LP forfeits the natural spread that could have been earned from benign, uninformed flow. The revenue lost is the standard bid-ask edge.
- **Under-Externalization (θ is too high):** When $\theta \to 1$, the LP is overconfident and internalizes almost all trades, effectively disabling the risk management system. While the LP captures the spread upfront, they absorb massive toxic flow from informed traders (like Clients E and F), leading to severe adverse selection where the mid-price systematically drifts against the LP's inventory.

The optimal threshold θ^* isolates the exact probability inflection point where the expected cost of adverse selection perfectly equals the expected spread capture.

3. Global vs. Client-Specific Thresholds

Theoretically, **client-specific thresholds** should strictly dominate a global threshold. Because clients exhibit vastly different toxicity profiles (as proven in Tasks 1 and 2), forcing a single global θ means we are likely over-externalizing safe clients (A, B) and under-externalizing toxic clients (E, F).

This theoretical advantage is visible in the validation and optimal parameter data. For instance, at $\tau = 30$, the optimizer assigned $\theta^* = 0.5540$ and 0.5910 to Clients A and B (allowing high internalization), while aggressively setting $\theta^* = 0.000$ for Clients E and F (forcing 100% externalization of their toxic flow). Consequently, the Client-Specific approach achieved a higher Total Validation PnL across all horizons.

The Overfitting Reality: However, analyzing the out-of-sample Test PnL reveals a different empirical reality. In four out of the six time horizons ($\tau \in \{10, 15, 20, 25\}$), the **Global threshold outperformed the Client-Specific thresholds on the Test set**. For example, at $\tau = 25$, the Global Test PnL was \$40,207.53 compared to the Client-Specific Test PnL of \$38,926.03.

Optimizing six independent thresholds on a limited validation set introduces excessive degrees of freedom, leading to variance and overfitting. While client-specific thresholds are structurally superior, a **global threshold acts as a powerful regularizer**, yielding a more robust out-of-sample performance given the current dataset size.

4. Final Test Set PnL Summary

The table below summarizes the final out-of-sample Test PnL for both the Global optimization and the aggregated Client-Specific optimization across all evaluated time horizons.

Horizon (τ)	**Global Test PnL**	**Client-Specific Test PnL**	**Best Approach (Out-of-Sample)**
5 seconds	\$37,243.11	\$37,243.11	Tie
10 seconds	**\$34,327.28**	\$33,865.65	Global
15 seconds	**\$34,791.35**	\$34,711.26	Global
20 seconds	**\$35,973.01**	\$35,926.59	Global
25 seconds	**\$40,207.53**	\$38,926.03	Global
30 seconds	\$42,656.42	**\$42,929.32**	Client-Specific